

box size
(geometry)detection
confidencelocalization
qualitydetector σ
(NMS / DFL)track-side
history

cost: additive

ours -3.75

cost: multiplicative

ByteTrack impl
(no rationale)

LG-Track eq. 1, 3

ours -1.01

distance function

NWD (detection)
ours -4.98*ours* -8.90
*loses to size by 3.93*Kalman R UCMCTrack
 $R = (\sigma_m w, \sigma_m h)$ NSA Kalman
 $\tilde{R} = (1 - c)R$ UTrack -0.62
UncTrack -0.1

gating / box expansion

ours (control)
beats σ
*(criterion-dep.)*UncTrack +2.2
ours -4.33

association ordering

LG-Track
4-level cascadeUncTrack
entropy +0.1DeepSORT (age)
Bae (routing)

matching threshold

ours -0.21
(oracle +0.89)

italic = this work



gain reported



loss measured



neutral / baseline



mixed (different quantities)